

MANAGEMENT OF KIDNEY CYSTS **IN PRIMARY CARE**

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INTRODUCTION

Kidney cysts are fluid-filled sacs that develop within the kidneys. They are typically non-cancerous and can vary in size, ranging from very small cysts that are barely noticeable to large cysts that can cause discomfort and affect kidney function.

Epidemiology

Benign kidney cysts are common, with an estimated 30% of patients above 60 years old who will be diagnosed with at least one simple kidney cyst through abdominal imaging.

The prevalence of kidney cysts increases with age and is also higher in males than females.

Importance

In the primary care setting, kidney cysts are often diagnosed incidentally. It is important for primary care physicians to differentiate between benign and complex cysts as complex cysts are associated with an increased risk of malignancy. These types of cysts may require further imaging, biopsy, or surgery for diagnosis and management.

Kidney cysts can also be present due to other conditions such as autosomal dominant polycystic kidney disease, Von Hippel-Lindau syndrome, and prolonged haemodialysis in end-stage renal disease.

SCREENING AND DIAGNOSIS

In many cases, benign kidney cysts do not cause any symptoms and are discovered incidentally. However, larger cysts or cysts that multiply and enlarge over time may lead to symptoms such as:

- Flank or back pain
- Urinary frequency
- Haematuria
- Hypertension
- Urinary tract infections
- Kidney stones
- Chronic kidney disease in advanced cases

Diagnosis

Kidney cysts are commonly diagnosed through imaging tests, such as ultrasound kidneys, in the primary care setting. Other radiological investigations such as CT and MRI scans can visualize the size, number, and characteristics of these cysts.

Classification of Kidney Cysts

Kidney cysts detected on CT or MRI are classified based on the Bosniak classification system (2019). With the availability of US kidneys in primary care, we can risk stratify cysts for further imaging.

Type	US Appearance	Recommendation
Simple	- Thin smooth wall - Anechoic, no septa, calcification or solid component	No follow-up required, unless symptomatic
Complex	- Few thin septa - Septa and wall may appear echogenic - May have fine calcifications / milk of calcium cysts - Small cyst (< 6cm)	Look for previous US or CT to assess change in size over time If not repeat imaging in 1 year If stable, discuss with patient No follow-up required, unless symptomatic Risk of malignancy < 1%
	Thickened hyperechoic wall Thickened septa, or many thin septa Multiple, coarse calcium calcifications within cyst - Any one cyst ≥ 6cm	Refer Urology (Routine)
	≥ 1 hyperechoic thick or irregular walls or multiple thickened septa	Refer Urology (Early)
	- Solid nodule - Presence of vascularity inside index lesion (with Doppler)	Refer Urology (Direct Access)

MANAGEMENT & REFERRAL

Recommended Follow-Up Interval

1. **Simple** renal cysts are benign and do not require any further follow-up imaging.
2. **Complex** cysts, large cyst size, multiple cysts, solid nodule, thick septa, multiple calcification should be referred to Urology for further imaging.

Other Indications for Referral to Urology

1. Patients with symptomatic kidney cysts
2. Patients with multiple cysts and family history suggestive of other conditions such as polycystic kidney disease, Von Hippel-Lindau syndrome
3. Incidental findings of hydronephrosis, **stone**, or **suspected** tumour must be referred **urgently to Urology (direct access)**

Indications for Referral to Emergency Department

1. Cyst infection / rupture with symptoms and signs including but not limited to, fever, flank pain, haematuria, hypotension
2. **Hydronephrosis, stone or tumour with symptoms and signs of urosepsis or severe renal impairment**

Things to Note

1. Radiological results labelled as "R1U - Unexpected" demand additional attention and should not be overlooked or dismissed
 - a. Example: "Tiny calcification in the left mid kidney about 5mm with no hydronephrosis.
Echogenic nodule in the left mid kidney about 24 x 22 x 22mm with small eccentric cystic component and some vascularity.
Comments:
Fatty change in the liver and tiny right hepatic cyst.
Echogenic nodule in the left mid kidney. Further evaluation suggested.
Likely tiny left mid renal stone"
 - b. The above report suggests the possibility of renal cell carcinoma; referral is warranted.
2. For incidental renal cysts identified during US HBS where the complexity could not be determined, further US kidneys is recommended

ABBREVIATIONS

CT	Computed tomography
HBS	Hepatobiliary System
MRI	Magnetic resonance imaging
US	Ultrasound

REFERENCES

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3. Richard, P. O., et al (2023). CUA GUIDELINE. *CUAJ*, 17(6).
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APPENDIX

Bosniak (CT) Classification of Kidney Cysts

Bosniak Class	US Appearance	Risk of Malignancy (%)	Recommendation
I	- Thin smooth wall ($\leq 2\text{mm}$) which may enhance - Homogenous simple fluid - No septa, calcification or solid component	0%	No follow-up required, unless symptomatic
II	- Thin smooth wall ($\leq 2\text{mm}$) which may enhance - Few hairline thin septa ($\leq 2\text{mm}$) - Septa and wall may enhance - May have fine calcifications - Small hyperdense cysts ($< 6\text{cm}$)	1%	Look for previous US or CT to assess change in size over time If not, repeat imaging in 1 year If stable, discuss with patient No follow-up required, unless symptomatic
II-F	- Minimally thickened (3mm) enhancing wall, or smooth minimal thickening (3mm) of one or more enhancing septa, or many (≥ 4) hairline thin septa ($\leq 2\text{mm}$) • Multiple coarse calcifications within cysts - Large hyperdense cysts ($\geq 6\text{cm}$)	1 – 38%	Refer Urology (routine)
III	One or more enhancing thick ($\geq 4\text{mm}$) or enhancing irregular walls or septa	50%	Refer Urology (early)
IV	One or more enhancing nodule(s)	$> 90\%$	Refer Urology (direct access)